

Sulzer CPT-50 Centrifugal Pump

Operation, Maintenance & Troubleshooting Manual

Applicable Units: P-03, P-04, P-08, P-11 (Naphtha / Kerosene Service)

1. Pump Specifications

Parameter	Value
Model	CPT-50
Type	Single-stage, between-bearings, centrifugal
Service fluid	Naphtha / light hydrocarbons / kerosene
Design flow	80 m ³ /hr
Design head	60 m
Rated power	22 kW
Rated speed	2,950 RPM
Impeller diameter	240 mm
Impeller type	Semi-open, 4-vane
Casing material	316L stainless steel (naphtha service)
Impeller material	Duplex stainless steel 2205
Shaft material	316 stainless steel
Bearing type	FAG 6308 (both ends)
Seal type	Mechanical single seal, API Plan 11
NPSH required	2.8 m at design flow
Maximum allowable temperature	120°C
Maximum allowable pressure	10 bar
ATEX classification	II 2G Ex d IIB T3 (naphtha service)

2. Normal Operating Envelope

2.1 Flow & Head Envelope

- Minimum continuous flow: 32 m³/hr (40% of BEP)
- Best Efficiency Point (BEP): 80 m³/hr at 60 m head
- Maximum continuous flow: 92 m³/hr (115% of BEP)

CAUTION — Naphtha Service: Operating below minimum flow causes rapid temperature rise in the pump casing. Naphtha flash point is 38°C — overheating creates fire and explosion risk. Always maintain minimum flow or open recirculation line.

2.2 Bearing Temperature Limits

Condition	Bearing Temp (°C)	Action
Normal operation	40–60°C	No action required
Elevated — monitor	60–75°C	Check lubrication; increase monitoring
Warning threshold	75–80°C	Inspect within 4 hours
Alarm threshold	>80°C	Reduce load; inspect immediately
Failure threshold	>90°C	Immediate shutdown

Note for naphtha service: Lower alarm thresholds than crude service due to lower fluid flash point and lighter bearing loads.

2.3 Vibration Limits (ISO 10816-3, Group 2)

Zone	Vibration (mm/s RMS)	Condition	Action
A	0 – 2.3	New or recently overhauled	Normal
B	2.3 – 4.5	Acceptable long-term	Monitor
C	4.5 – 7.1	Unsatisfactory	Plan maintenance
D	> 7.1	Dangerous	Shutdown immediately

Axial vibration note: CPT-50 semi-open impeller is sensitive to axial thrust. Axial vibration >3.5 mm/s RMS indicates impeller-to-casing clearance issue or misalignment — inspect within 48 hours.

2.4 Suction Pressure & Cavitation

- Minimum suction pressure: 1.5 bar absolute (to prevent cavitation)
- Cavitation indicators: crackling/popping noise, vibration increase, flow instability

- If suction pressure <1.2 bar: shutdown risk — check suction line for blockage or vapor lock
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3. Bearing Maintenance

3.1 Lubrication Specifications

- Lubricant: FAG Arcanol LOAD150 grease (NLGI Grade 2)
- Alternative: Shell Gadus S2 V220
- Relubrication interval: every 2,500 operating hours
- Grease quantity: 12 g per bearing
- **Do not mix grease types** — incompatible thickeners cause bearing failure

3.2 Bearing Replacement Criteria

- Temperature consistently >75°C despite fresh lubrication
- Vibration Zone C or D
- Audible noise: grinding or irregular knocking
- After any temperature excursion >90°C
- Scheduled replacement: every 25,000 operating hours or 3 years

3.3 FAG 6308 Bearing Data

- Dynamic load rating (C): 41.0 kN
- Static load rating (C0): 24.0 kN
- Limiting speed (grease): 7,500 RPM
- L10 life at design conditions ($P \approx 4.2$ kN, $n = 2,950$ RPM): ~38,000 hours

3.4 Bearing Replacement Procedure

1. Isolate pump; purge naphtha from casing (nitrogen purge — minimum 3 volumes)
 2. Verify LEL <10% before opening any flanges (gas detector required)
 3. Remove bearing housing end covers
 4. Extract shaft; remove bearings using hydraulic puller
 5. Clean shaft journal; inspect for scoring (replace shaft if scoring >0.05 mm deep)
 6. Heat new FAG 6308 to 80°C using induction heater; press onto shaft
 7. Allow to cool; verify zero axial play
 8. Repack with 12 g Arcanol LOAD150
 9. Reassemble; align to ≤ 0.05 mm TIR
 10. Nitrogen purge casing before introducing naphtha
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4. Mechanical Seal Maintenance

4.1 Seal System — API Plan 11 (Flush from Discharge)

- Flush flow: 2–4 L/min from pump discharge to seal chamber
- Flush line orifice: 3 mm (do not enlarge — controls flush rate)
- Seal chamber pressure: approximately 0.5 bar above suction pressure
- Seal face material: Silicon carbide vs. carbon graphite

4.2 Seal Failure Indicators

- Visible naphtha leakage at seal housing: immediate shutdown (fire hazard)
- Seal leak proximity sensor activated: shutdown within 5 minutes
- Flush line flow loss: check orifice for plugging
- Seal face temperature $>120^{\circ}\text{C}$: flush flow insufficient

4.3 Seal Replacement Procedure

SAFETY: Naphtha is highly flammable. Full PPE required. No ignition sources within 10 m. 1. Shutdown pump; close suction and discharge isolation valves 2. Depressurize casing; nitrogen purge (3 volumes minimum) 3. Verify LEL $<10\%$ with gas detector 4. Drain seal chamber via drain plug 5. Remove gland plate bolts ($4 \times \text{M12}$, torque $35 \text{ N}\cdot\text{m}$) 6. Slide seal off shaft; inspect faces and O-rings 7. Install new seal; verify face flatness (max 0.001 mm deviation) 8. Torque gland plate bolts evenly in cross pattern 9. Pressurize flush system; check for leaks before startup 10. Typical replacement time: 3–4 hours

4.4 Seal Life Expectancy

- Normal naphtha service: 12–18 months
 - Reduced life factors: dry running, particulate contamination, temperature $>100^{\circ}\text{C}$, shaft runout $>0.05 \text{ mm}$
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5. Semi-Open Impeller Clearance Setting

5.1 Importance of Impeller Clearance

The CPT-50 uses a semi-open impeller. The axial clearance between impeller vane tips and the casing wear plate directly affects efficiency and vibration: - Too tight ($<0.3 \text{ mm}$): impeller rubs casing — vibration, heat, damage - Optimal ($0.3\text{--}0.5 \text{ mm}$): best efficiency, lowest vibration - Too loose ($>0.8 \text{ mm}$): efficiency loss $>8\%$, increased recirculation

5.2 Clearance Adjustment Procedure

1. Loosen bearing housing lock nuts

2. Advance shaft toward casing until impeller just contacts wear plate (zero clearance)
3. Back off shaft 0.4 mm using adjustment screws (1 turn = 0.5 mm on M10 thread)
4. Lock bearing housing; verify clearance with feeler gauge through inspection port
5. Rotate shaft by hand — must turn freely with no contact

5.3 Wear Plate Replacement

- Replace wear plate when: clearance cannot be reduced below 0.8 mm by adjustment
- Wear plate material: Duplex 2205 (same as impeller)
- Replacement interval: typically 3–5 years in naphtha service

6. Troubleshooting Guide

Symptom	Probable Cause	Corrective Action
Bearing temp >75°C	Insufficient lubrication	Re-grease with Arcanol LOAD150
Bearing temp >75°C	Misalignment	Check coupling alignment
Axial vibration >3.5 mm/s	Impeller clearance too tight	Adjust clearance to 0.4 mm
Axial vibration >3.5 mm/s	Angular misalignment	Realign coupling
Cavitation noise	Low suction pressure	Check suction line; increase NPSH
Flow declining	Impeller clearance too large	Adjust clearance; replace wear plate
Seal leak	Seal face damage	Replace seal immediately (fire hazard)
Motor overcurrent	Impeller rubbing	Check and adjust clearance
Pump overheating	Below minimum flow	Open recirculation line